1. Pipeline（工作流程）
该研究提出了一个结合大语言模型（LLM）生成和神经排名模型（NRM）评估的多阶段框架，旨在构建高质量的对抗性数据集

检索上下文构建： 针对特定查询，使用 神经排名模型（NRM） 检索出排名靠前的文档作为上下文基础，模拟真实的搜索环境

对抗句子生成（Adversarial Sentence Generation）： 利用 RAG（检索增强生成）技术，提示 LLM（如 Qwen3）结合查询、目标文档和上下文信息，生成满足流畅性、间接相关性（不直接回答查询）的对抗句

排名评估： 将生成的句子插入文档，使用目标 NRM（如 Cross-Encoder）计算其在排名中的位置变动（Ranking Delta）

NRM 评估与约束检查： 使用目标 NRM 评估变体的排名提升，并检查其是否满足三个核心约束：(O1) 流畅性与连贯性、(O2) 间接相关性（即不直接回答查询）、(O3) 排名提升

迭代自细化循环（Self-Refinement）： 如果初始生成未达到预设的排名提升目标（如进入前 10 或前 50），系统会将排名信号作为奖励反馈给 LLM，引导其调整策略并重新生成，直到满足攻击强度或达到最大迭代次数

2. contribution

提出了一个 “自我细化（Self-refinement）” 的反馈循环。它利用 NRM 的排名变化作为奖励信号，引导 LLM 迭代修改对抗性句子，直到成功提升排名

自细化生成框架： 提出了一种基于思维链（CoT）和排名反馈的自适应攻击模型，能够针对黑盒排名器自动发现漏洞

3. 这篇文章的思路

利用 RAG 提升对抗质量： 与以往的启发式扰动不同，该文认为对抗攻击应置于真实的检索上下文中。通过 LLM 理解“为什么其他文档排在前面”，生成的干扰项才更具威胁性

通过反馈实现“黑盒优化”： 核心思想是将对抗攻击建模为一种近似策略迭代。LLM 就像一个不断调整策略的代理，通过观察 NRM 反馈的排名这一非线性信号，逐步摸索出提升排名的规律

平衡攻击力与隐蔽性： 该研究强调对抗性不仅仅是排名的提升，更重要的是通过“语义间接相关”来保持文档的自然流畅，从而在避开人工审核和语言模型连贯性检查的同时实现攻击目标

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SHORT-PAPER

Datasets for Supervised Adversarial Attacks on Neural Rankers

AMIR KHOSROJERDI, University of Toronto, Toronto, ON, Canada
AMIN BIGDELI, University of Waterloo, Waterloo, ON, Canada
RADIN HAMIDI RAD, University of Toronto, Toronto, ON, Canada
MORTEZA ZIHAYAT, Toronto Metropolitan University, Toronto, ON, Canada
CHARLES L A CLARKE, University of Waterloo, Waterloo, ON, Canada
EBRAHIM BAGHERI, University of Toronto, Toronto, ON, Canada

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Datasets for Supervised Adversarial Attacks on Neural Rankers

Amir Khosrojerdi*
amir.khosrojerdi@mail.utoronto.ca
University of Toronto
Toronto, Ontario, Canada

Amin Bigdeli*
abigdeli@uwaterloo.ca
University of Waterloo
Waterloo, Ontario, Canada

Radin Hamidi Rad
radin.rad@utoronto.ca
University of Toronto
Toronto, Ontario, Canada

Morteza Zihayat
mzihayat@torontomu.ca
Toronto Metropolitan University
Toronto, Ontario, Canada

Charles L. A. Clarke
claclark@gmail.com
University of Waterloo
Waterloo, Ontario, Canada

Ebrahim Bagheri
ebrahim.bagheri@utoronto.ca
University of Toronto
Toronto, Ontario, Canada

Abstract

We introduce a novel dataset for *adversarial rank attacks against neural rankers*, enabling systematic research on robustness. Unlike prior unsupervised or surrogate-based methods, our approach uses Retrieval-Augmented Generation (RAG) with a Large Language Model (LLM) to create high-quality adversarial examples that subtly alter rankings while maintaining coherence and relevance. Built via a self-refining LLM-Ranker feedback loop, the dataset includes two tiers: *Gold* and *Diamond*, based on attack strength, along with rich metadata, ranking labels, and quality metrics. Released with code and prompts, it supports training, evaluation, and benchmarking of robust ranking systems.

CCS Concepts

• **Information systems → Adversarial retrieval**.

Keywords

Adversarial information retrieval, Retrieval robustness, Large language models, Retrieval-augmented generation

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1 Introduction

Neural Ranking Models (NRMs) have demonstrated state-of-the-art performance in Information Retrieval (IR) tasks by leveraging deep contextual representations and large-scale training corpora [11, 18]. Nevertheless, a growing body of work has identified that NRMs are susceptible to adversarial attacks, including document perturbations or injections that are imperceptible or semantically marginal yet induce significant changes in the model's output ranking [1, 2, 4, 12, 23, 25]. The manipulation of retrieval rankings via adversarial content has raised substantial concerns not only in academic settings but also in deployed systems, where such vulnerabilities may be exploited for adversarial search engine optimization (SEO), visibility manipulation, dissemination of low-quality content, or text-attributed systems manipulation [3, 6, 9, 15, 16, 20, 21]. These observations motivate a systematic and replicable investigation into the mechanisms and limitations of ranking robustness [5, 13] under adversarial conditions.

From a theoretical standpoint, adversarial vulnerability in NRMs stems from the complex and non-linear relationship between input representations and learned ranking functions [11, 14, 19, 28]. This is especially pronounced in neural architectures that operate in high-dimensional semantic spaces, where small shifts in document embeddings can yield disproportionate changes in similarity scores [8, 18]. In adversarial settings, this phenomenon is often exploited by introducing content that is not directly relevant to the query but contributes latent semantic features aligned with the learned notion of relevance. Crucially, to distinguish adversarial manipulation from legitimate content augmentation, it becomes necessary to formalize a *notion of indirect relevance*, wherein added text increases ranking scores without addressing the query explicitly. This requirement introduces a semantic constraint that prior adversarial IR methods typically ignore, as most rely on syntactic or heuristic perturbations, trigger-based augmentations, or query-injection techniques that trivially improve rank by increasing lexical overlap [12, 23].

While existing work has introduced attack strategies for NRMs, these approaches are often grounded in unsupervised optimization or reinforcement learning with surrogate models [1, 4]. The absence of datasets with supervision over adversarial impact and quality has limited the reproducibility, benchmarking, and theoretical analysis of ranking attacks. More fundamentally, without access to controlled adversarial corpora, it is difficult to analyze the generalization behavior of NRMs under perturbed conditions or to quantify the inductive biases that make them susceptible to particular adversarial patterns.

To address this gap, we present a principled framework for constructing retrieval-conditioned adversarial datasets, grounded in supervised generation and guided by formal constraints on fluency, coherence, and indirect relevance. Our approach leverages Retrieval-Augmented Generation (RAG) [10], wherein a Large Language Model (LLM) is prompted with a query, a target document, and a retrieval context of top-ranked documents. This setting mimics the retrieval condition that occurs in ranking systems, and prompts the LLM to generate content that is topically coherent with the query but does not explicitly answer it. The use of a contextual LLM generator, rather than token-level perturbations, ensures that

*Both authors contributed equally to this research.

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generated sentences are fluent and plausibly embedded in natural discourse [7, 27].

To formalize our generation criteria, we define three objectives: (1) the adversarial modification must preserve fluency and coherence with the document's style and structure; (2) the added content must exhibit indirect relevance, avoiding direct entailment or paraphrasing of the query; and (3) the modified document must exhibit a measurable improvement in rank under a given NRM. These constraints enable a rigorous definition of adversariality in the ranking context, one that goes beyond lexical overlap or syntactic mutation and aligns with the epistemic and operational semantics of modern retrieval systems.

Furthermore, we introduce a self-refinement mechanism that implements an iterative feedback loop between the LLM and the neural ranker. This loop serves two purposes. First, it increases the likelihood that generated content satisfies the ranking constraint by providing the LLM with reward signals based on actual retrieval outcomes. Second, it enables an implicit approximation of gradient-based optimization over black-box rankers by conditioning future generations on previously successful perturbations. The iterative refinement structure can be viewed as an application of approximate policy iteration, where the LLM updates its generative policy based on reward signals from the ranking environment.

We instantiate our framework as a dataset and codebase with two variants: *Gold*, containing the most effective adversarial sentence per query-document pair, and *Diamond*, which applies stricter ranking thresholds to capture high-impact attacks. Each entry includes the query, document, insertion point, ranking delta, fluency (via perplexity), and syntactic acceptability metrics, supporting detailed evaluation and comparison. Built on the MS MARCO passage ranking benchmark [17] using a target NRM, the release includes all prompts, refinement code, and evaluation scripts for transparency and reproducibility.

The primary contributions of this work are threefold. **First**, we introduce the first supervised, retrieval-conditioned dataset for adversarial document ranking that satisfies formal semantic and ranking constraints. **Second**, we provide a complete generation and evaluation pipeline that includes a chain-of-thought refinement mechanism grounded in observable ranking signals. **Third**, we deliver a reusable and extensible resource for the community, supporting downstream research in adversarial IR, robust ranking model evaluation, retrieval-based LLM augmentation, and generative attack modeling.

2 Proposed Approach

This resource introduces a systematic approach for constructing adversarial document datasets that reveal NRMs vulnerabilities to subtle content manipulation under realistic retrieval settings. Each example is created by injecting a sentence into a target document to boost its ranking for a given query without adding obvious semantic artifacts. To ensure quality and adversarial validity, we enforce three constraints: (O_1) fluency and coherence with the original text; (O_2) indirect relevance that influences ranking without directly satisfying the query; and (O_3) an improved rank under the target NRM.

2.1 Problem Definition

Let $\mathcal{Q} = \{q_1, q_2, \ldots, q_n\}$ denote n user queries submitted to an IR system. For each query $q \in \mathcal{Q}$, the system returns a ranked list of documents $\mathcal{D}_q = \{d_1, d_2, \ldots, d_m\}$, where the ranking is determined by a scoring function $f_{\text{rel}} : \mathcal{Q} \times \mathcal{D} \to \mathbb{R}$, such that:

$$\text{rank}(q, d) = \sum_{d' \in \mathcal{D}_q} \mathbb{1}\left[f_{\text{rel}}(q, d') > f_{\text{rel}}(q, d)\right] \tag{1}$$

Given a query q, a target document $d_t \in \mathcal{D}_q$, and a rank threshold $k \in \mathbb{N}$, the goal is to construct a perturbed version d_t^{adv} of the document such that $\text{rank}(q, d_t^{\text{adv}}) \leq k$, subject to the required objectives (O_1–O_3). The problem is to find such a transformation function $\phi : \mathcal{D} \times \mathcal{Q} \to \mathcal{D}^{\text{adv}}$ that satisfies the above ranking and semantic constraints for a given retrieval model f_{rel}.

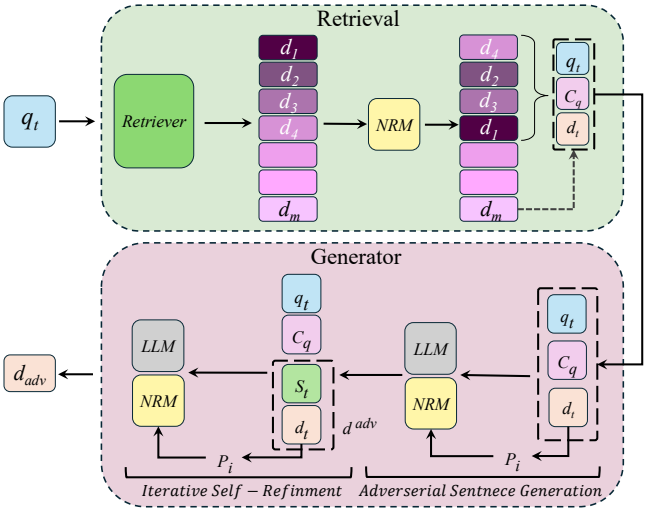

Figure 1: Overview of our proposed workflow.

2.2 Approach Overview

Figure 1 illustrates our multi-stage pipeline combining LLM generation and NRM evaluation. Given a query, we retrieve top-ranked context documents and prompt the LLM with the query, target document, and context to generate adversarial candidates (**adversarial sentence generation**). These are inserted at various positions to create perturbed variants and ranked by the NRM. Variants meeting all constraints are retained. Through a **self-refinement loop**, the LLM conditions new generations on prior successful examples, enhancing perturbation effectiveness and diversity. Each dataset instance includes the original and perturbed documents, insertion point, ranking delta, and fluency and acceptability metrics, supporting training and evaluation of adversarial IR models. We formalize this process below.

2.3 Adversarial Dataset Construction

Given a query $q \in \mathcal{Q}$, a target document $d_t \in \mathcal{D}_q$, and a set of top-c context documents $\mathcal{C}_q = \{d_1, d_2, \ldots, d_c\} \subset \mathcal{D}_q$, we define an adversarial sentence generator $\mathcal{G} : \mathcal{Q} \times \mathcal{D} \times \mathcal{C} \to \mathcal{S}$, which outputs a set of candidate adversarial sentences $\mathcal{S} = \{s_1, s_2, \ldots, s_l\}$. Each candidate sentence $s \in \mathcal{S}$ is inserted into the original target document d_t at position $p \in \mathcal{P}(d_t)$, where $\mathcal{P}(d_t)$ denotes the set of allowable insertion points (e.g., sentence boundaries), yielding a perturbed document $d_{t(p,s)}^{\text{adv}} = \text{Insert}(d_t, s, p)$. We include $d_{t(p,s)}^{\text{adv}}$ in the adversarial dataset $\mathcal{D}_{\text{adv}}$ if and only if the following constraints are satisfied: **(a)** *Ranking Improvement*: $\text{rank}(q, d_{t(p,s)}^{\text{adv}}) \leq \text{rank}(q, d)$;

(b) *Indirect Relevance*: $s \not\models q$, i.e., the sentence does not explicitly or implicitly answer the query; **(c)** *Linguistic Coherence*: $d^{\text{adv}}_{t(p,s)}$ preserves the fluency and semantic consistency of d_t, as judged by a coherence function $\psi(d^{\text{adv}}_{t(p,s)}) > \tau$ for threshold τ. This process yields a dataset $\mathcal{T} = \{(q_i, C_{q_i}, d_t, s, p)\}_{i=1}^{N}$ of adversarial training examples, where each s_i is a validated perturbation at position p that satisfies constraints (a), (b) and (c).

To construct this dataset, we implement a generation pipeline that combines an LLM and a NRM in an iterative loop. The LLM $\mathcal{G}(q, d_t, C_q)$ proposes candidate adversarial sentences based on a query q, a target document d_t, and the set of context documents C_q. Each sentence is inserted into the target document and evaluated by the NRM $\mathcal{R}(q, d^{\text{adv}}_{t(p,s)})$, which measures its impact on the document's ranking. If a candidate yields a valid rank improvement within the k threshold, while preserving fluency and semantics, it's retained; otherwise, the LLM enters a refinement phase, using ranking feedback to improve its generations.

Adversarial Sentence Generation. Given the query q, our approach first constructs the contextual basis for the LLM generator $\mathcal{G}$ by selecting the top-c ranked documents as the context set $C_q = \{d_1, d_2, \ldots, d_c\}$, where c represents the number of documents incorporated as context for adversarial sentence generation. In order to generate adversarial sentences, we prompt the LLM to produce an initial set $\mathcal{S}_t = \mathcal{G}(q, d_t, C_q)$. Each generated sentence $s \in \mathcal{S}_t$ is then inserted at position p, creating an adversarially modified document.

Next, the NRM model $\mathcal{R}$ evaluates the ranking impact of the adversarial sentence being injected into the document as follows:

$$\text{rank}(q, d^{\text{adv}}_{t(p,s)}) = \mathcal{R}(q, d^{\text{adv}}_{t(p,s)}) \tag{2}$$

If the modified document achieves rank $\text{rank}(q, d^{\text{adv}}_{t(p,s)})$, the process terminates; otherwise, the system enters the feedback loop.

Iterative Self-Refinement. If no adversarial sentence yields the desired ranking improvement, the system refines the sentences iteratively. For this purpose, the NRM component selects the most effective adversarial sentences:

$$\mathcal{S}^{p,\text{high}}_t = \arg\min_{s \in \mathcal{S}^p_t} \text{rank}(q, d^{\text{adv}}_{t(p,s)}) \tag{3}$$

The most effective adversarial sentences from the previous step, $\mathcal{S}^{p,\text{high}}_t$, are used to guide the language model in generating refined candidates. These are then inserted into the original document d_t at position p to produce new adversarial versions. Each updated document is evaluated by the neural ranking model (NRM) to assess its impact on the query-document ranking. This process repeats until the adversarial document reaches a top-k rank or the maximum number of iterations is reached.

3 Implementation Setup

All dataset variations along with the code and LLM prompts are publicly available in our GitHub repository[1].

Model Details: To build our RAG-based adversarial dataset, we use Qwen3 32B [26], selected due to its strong language understanding, context-aware generation, and fluency preservation. The model is prompted with Target Query, Target Document, and Top-ranked Documents retrieved via a NRM. We select `msmarco-MiniLM-L-12-v2` cross encoder [22] for ranking evaluation. We set the maximum number of refinement iterations to 5, selected the top-5 ranked documents as the context, and stopped refinement once an adversarial document reached the top-10 (k=10).

Establishing Benchmarks: We evaluate our datasets against strong baselines across four attack types: word-level, trigger-based, sentence-level, and prompt-based. **(1)** `PRADA` [25] replaces up to 20 key tokens with synonyms using a surrogate ranker. **(2)** `Brittle-BERT` [23] prepends up to 12 trigger tokens to the document. **(3)** `PAT` [12] inserts trigger words at the beginning based on surrogate model cues. **(4)** `IDEM` [4] injects up to 500 BART-generated connector sentences. **(5)** `EMPRA` [1] perturbs sentence embeddings to align with query-relevant semantics while preserving meaning. **Evaluation Metrics:** Following prior work [1, 4, 25], we assess attack effectiveness and text quality based on several metrics, namely *Attack Success Rate (ASR)* is the percentage of documents with rank improvement. *Boosted Top-10* and *Boosted Top-50* report the proportion of documents reaching rank ≤ 10 or 50, respectively. *Boost* captures average rank improvement. *Perplexity (PPL)* (via `GPT-2`) evaluates fluency; lower is better. *Acceptability Score (AcS)* uses a neural classifier [24] to score the acceptability of text.

Source Data: Consistent with prior studies [1, 4, 12, 23, 25], we establish the benchmarks using the MS MARCO passage V1 collection [17]. Building the foundation of our proposed datasets on this large-scale, and well-annotated corpus, we deliver an adversarial attack dataset suited for evaluating IR system robustness. Following the approach of [1, 4], we randomly sampled subset of 1,000 queries from the MS MARCO dev small. For each query, we randomly select two distinct types of documents, Easy-5 and Hard-5, selected from the victim NRM's re-ranking of the top-1 K BM25 candidates. *Easy-5* documents consists of five documents initially ranked between positions 51 and 100, sampled at equal intervals (e.g., ranks 51, 63, 76, 84, 91) to assess mid-rank improvements. In contrast, *Hard-5* group comprises the five lowest-ranked documents, representing the most challenging cases for rank boosting. For each query–document pair, we release a RAG-based adversarially augmented dataset.

Dataset Variants: To cover diverse augmentation scenarios, we define two dataset variants based on rank improvement. *(1) Gold:* For each query-document pair, we select the best-performing sentence across insertion positions, capturing optimal augmentation cases. *(2) Diamond:* Includes only instances where the perturbed document meets strict ranking thresholds, final rank ≤10 for Easy-5, and ≤50 for Easy-5 and Hard-5, highlighting the most impactful attacks.

4 Dataset Evaluation

We evaluate our datasets to demonstrate their utility as practical and rigorous benchmarks for adversarial IR. The Gold and Diamond variants are constructed to reflect various realistic adversarial scenarios, enabling researchers to study the impact of document perturbations across varying levels of ranking difficulty and insertion strategies.

Gold Dataset Performance. The Gold dataset offers a curated collection of adversarial augmentations that satisfy all three constraints—fluency, indirect relevance, and rank improvement—for each query–document pair. As shown in Table 1, it achieves a nearly perfect Attack Success Rate (ASR) across both Easy-5 and Hard-5 scenarios. This means the RAG-based generation method consistently produces effective, query-conditioned perturbations. In Easy-5

[1] https://github.com/KhosrojerdiA/adversarial-datasets

Table 1: Gold Dataset statistics over Easy-5 and Hard-5.

Group	Type	SentPos	Counts	ASR	%r ≤ 10	%r ≤ 50	Boost	PPL	AcS
Easy-5	Orig.	–	5,000	–	–	–	–	37.3	0.78
	Adv.	all	5,000	99.7	62.2	93.1	59.2	42.4	0.77
		v=0	3,134	100.0	72.9	96.6	63.2	42.2	0.77
		v=1	961	99.7	54.4	92.5	57.7	42.7	0.77
		v=2	371	99.7	44.5	87.9	51.6	40.0	0.76
		v=3	163	99.4	33.1	85.9	50.6	40.4	0.73
		v≥4	371	98.1	22.4	72.5	40.5	46.5	0.77
Hard-5	Orig.	–	5,000	–	–	–	–	51.4	0.72
	Adv.	all	5,000	99.9	29.1	49.4	781.1	67.4	0.72
		v=0	3,281	99.9	34.5	55.0	802.5	66.9	0.73
		v=1	854	100.0	25.3	44.7	773.3	67.2	0.69
		v=2	341	100.0	14.1	36.1	726.9	66.1	0.70
		v=3	142	100.0	7.7	28.9	701.6	64.6	0.63
		v≥4	382	100.0	12.8	31.2	692.3	74.0	0.72

Table 2: Diamond Dataset statistics over Easy-5 and Hard-5.

Group	Type	SentPos	Counts	%r ≤ 10	Boost	PPL	AcS
Easy-5	Orig.	—	3,110	–	–	37.6	0.78
	Adv.	all	3,110	100.0	69.7	41.9	0.77
		v=0	2,285	100.0	69.7	41.3	0.78
		v=1	523	100.0	69.7	44.8	0.76
		v=2	165	100.0	70.5	39.3	0.74
		v=3	54	100.0	67.3	38.4	0.75
		v≥4	83	100.0	70.2	48.7	0.74
Hard-5	Orig.	—	2,470	–	–	55.8	0.71
	Adv.	all	2,470	58.9	983.1	73.1	0.71
		v=0	1,805	62.7	984.0	70.8	0.72
		v=1	382	56.5	982.3	79.9	0.66
		v=2	123	39.0	979.3	74.8	0.71
		v=3	41	26.8	974.9	76.2	0.63
		v≥4	119	41.2	979.3	82.0	0.69

group, 62.2% of documents are boosted into the top-10 (%r ≤ 10) and 93.1% into the top-50 (%r ≤ 50), with an average boost of 59.2 rank positions. In Hard-5, 29.1% of adversarial documents reach the top-10 and 49.4% the top-50, with an average boost of 781.1 ranks.

We further explore the impact of Sentence Position (SentPos) of adversarial insertions. For Easy-5, injecting adversaries at the first position yields a perfect 100% ASR, with 72.9% of documents boosted into the top-10 and a rank boost of 63.2. The attack effectiveness reduces slightly at later positions. Documents modified at positions four or beyond achieve 98.1% ASR, delivering a smaller rank improvement (40.5).

In addition to significant rank improvements, the linguistic quality of the adversarial augmentations remains high. Comparing adversarial (Adv.) documents with their original counterparts (Orig.), we can see that perplexity increases slightly from 37.3 to 42.4 on Easy-5 and from 51.4 to 67.4 on Hard-5, indicating slightly higher uncertainty but still within acceptable ranges. Also, acceptability score (AcS) remain at a comparable level. Easy-5 experience AcS drop only from 0.78 to 0.77, and Hard-5 holds at 0.72, indicating that adversarial sentences continue to be perceived as fluent and well-formed text.

Diamond Dataset Performance. The Diamond dataset offers an even more selective view of adversarial effectiveness by including only those perturbations that exceed strict rank-improvement thresholds. As detailed in Table 2, the dataset isolates cases where adversarial content leads to substantial ranking jumps, making it a strong resource for stress-testing ranking models and examining worst-case vulnerabilities. Overall, there are 3,110 query–document pairs in the Easy-5 group, for which augmentations achieve 100% boosted top 10 with a mean boost of 69.7 ranks. Hard-5 remains more challenging as only 2,470 of augmentations achieve 100% boosted top 50 with 58.9% boosted top 10, and an average lift of 983.1 ranks. Exploring the insertion positions shows that for Easy-5, inserts at the beginning achieve the strongest results (69.7 rank boost), with only minor variation at deeper positions. In Hard-5, the earliest insertion delivers the best outcome (boosted top-10 of 62.7% and a 984.0 rank boost), while deeper positions see a gradual decline in performance.

Table 3: Performance comparison across baselines and ours.

Model	Easy-5					Hard-5				
	ASR	%r ≤ 10	%r ≤ 50	Boost	PPL	ASR	%r ≤ 10	%r ≤ 50	Boost	PPL
PRADA	76.9	2.6	44.7	22.1	96.8	75.7	0.0	2.2	98.5	202.8
Brittle-BERT	94.1	61.6	90.7	56.8	118.1	100.0	29.6	72.7	933.1	173.2
PAT	51.2	3.5	25.0	2.9	47.0	80.0	0.0	1.1	119.2	70.2
IDEM	99.3	82.4	97.4	67.0	38.8	99.9	45.5	80.2	926.3	61.8
EMPRA	99.7	87.9	99.2	69.3	37.3	99.4	51.3	80.9	904.8	55.0
Ours	100.0	100.0	100.0	69.7	41.9	100.0	58.9	100.0	983.1	73.1

Despite the remarkable effectivness, linguistic quality remains high. Easy-5 adversarial PPL increases modestly from 37.6 to 41.9, and Hard-5 PPL rises from 55.8 to 73.1, indicating slightly higher uncertainty by a fluency model yet still fluent text. Acceptability scores remain stable for Easy-5, decreasing slightly from 0.78 to 0.77, and for Hard-5 remaining at 0.71, demonstrating that even the most aggressive augmentations pass acceptability checks.

Comparative Evaluation. To demonstrate the utility of our proposed datasets, we conduct a comparative analysis using state-of-the-art adversarial attack methods evaluated on our Diamond dataset. The results, presented in Table 3, illustrate how our dataset enables robust and consistent evaluation across a diverse set of adversarial strategies. Overall, our Diamond dataset augmentations can achieve the highest attack performance metrics across both Easy-5 and Hard-5. Specifically, it achieves average boost of 69.7 for Easy-5 and 983.1 for Hard-5. While the strongest baseline, EMPRA, records a comparable Easy-5 boost of 69.3 and a slightly lower Hard-5 boost of 904.8, our Diamond's augmentations still achieves broader coverage by boosting every document into the top-10 in Easy-5 and into top-50 in both groups. In addition to strong effectiveness, our proposed dataset maintains competitive fluency by having the perplexity of 41.9 and 73.1 for Easy-5 and Hard-5, respectively. Although EMPRA and IDEM achieve lower perplexities, our dataset offers a well-balanced trade-off between attack effectiveness and fluency. This balance of high attack performance and low linguistic distortion underlines the strength of our retrieval-augmented generation framework for generating stealthy and effective adversarial texts.

5 Concluding Remarks

We present a structured and supervised resource for studying adversarial vulnerabilities in NRMs. Using adversarial sentence generation with iterative refinement and rank-based evaluation, our dataset enables fine-grained benchmarking, controlled training, and reproducible experimentation. The released data, codebase, and generation pipeline provide a practical foundation for advancing robustness research in information retrieval, supporting both attack modeling and defense evaluation. We anticipate this resource will facilitate new lines of inquiry into ranking resilience, adversarial generalization, and the interplay between generative models and retrieval systems.

6 Gen-AI Usage Disclosure

In compliance with the ACM authorship policy and CIKM 2025 guidelines, we hereby disclose our use of generative AI tools in the research process. Specifically, we employed the open-source generative language model Qwen3-32B [26] to generate adversarial sentences as part of our retrieval-augmented generation (RAG) based dataset construction pipeline. We also utilized this model to refine and optimize prompts used during the data generation process, ensuring more effective adversarial examples. Importantly, no generative AI was used to write the text of this manuscript; all writing and editing of the paper were done exclusively by the human authors without AI assistance. We affirm that our use of GenAI tools adheres to ACM's authorship and GenAI usage disclosure policies. The above details fully disclose how generative AI was utilized in our work, as required. We confirm that these practices are in line with the official guidelines and that the integrity of authorship for this paper remains uncompromised by our limited use of AI tools.

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